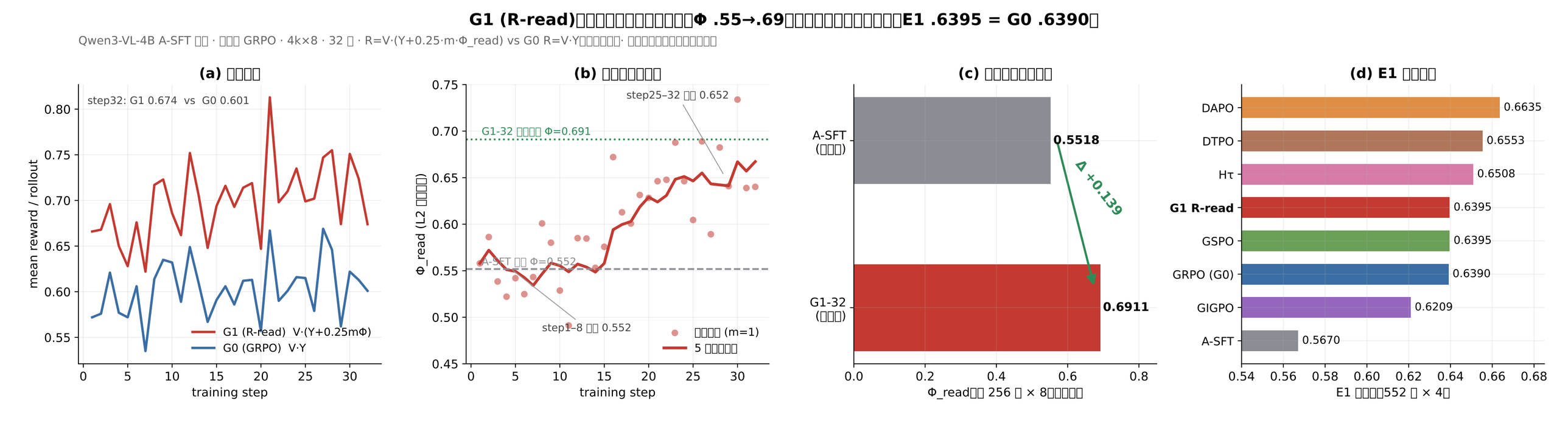
G1 (R-read)：Φ .55→.69，E1 .6395 = G0 .6390
Qwen3-VL-4B A-SFT · GRPO · 4k×8 · 32 · R=V·(Y+0.25·m·Φ_read) vs G0 R=V·Y ·
(a)
step32: G1 0.674 vs G0 0.601
mean reward / rollout
training step
G1 (R-read) V·(Y+0.25mΦ)
G0 (GRPO) V·Y
(b)
step25–32 0.652
G1-32 Φ=0.691
A-SFT Φ=0.552
step1–8 0.552
Φ_read (L2)
training step
(m=1)
5
(c)
A-SFT
G1-32
0.5518
0.6911
Δ +0.139
Φ_read 256 × 8
(d) E1
DAPO 0.6635
DTPO 0.6553
Hτ 0.6508
G1 R-read 0.6395
GSPO 0.6395
GRPO (G0) 0.6390
GIGPO 0.6209
A-SFT 0.5670
E1 552 × 4